# Ashmit Mukherjee

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New York University Abu Dhabi

## Education

**New York University Abu Dhabi** *Expected May 2027*
B.S. in Computer Science; minor in Economics. Cumulative GPA: 3.599/4.000
*Study away:* NYU Paris (Spring 2025); NYU New York (Fall 2025)
**Selected coursework:** Social Networks; Natural Language Processing; Principles of Data Science I; Statistics for the Social and Behavioral Sciences; Algorithms

## Research Interests

Computational social science; human-AI interaction; experimental methods; empirical machine learning.

## Research Experience

**Research Assistant, Human-AI Collaboration** *Feb 2026 - present*
NYU Abu Dhabi | With Ben Rosche and Hanan Salam

- Lead technical development and review of a faculty-led platform for studying AI advice and cooperation; contribute implementation-focused feedback on study design.
- Translate experimental rules and information boundaries into specifications for Python/FastAPI mechanics, Empirica participant workflows, session orchestration, and data exports.
- Coordinate checks of permitted actions, session recovery, and outcome calculations using scripted adviser inputs and independent reconstruction from recorded actions.

**Research Assistant, eBRAIN Lab** *Feb 2026 - present*
NYU Abu Dhabi | With Muhammad Shafique and Minghao Shao

- Extended protein-model evaluation to signal-peptide classification and cleavage-site prediction, including ESM-2 task adapters, model heads, and metrics.
- Ran LoRA, full fine-tuning, and biological-prior comparisons on the university HPC cluster across 150M, 650M, and 3B models; investigated initialization, feature injection, and efficiency as distinct experimental choices.
- In a three-seed 650M follow-up, measured 86.08% mean exact cleavage-site match for BLOSUM62 feature injection versus 83.64% for vanilla LoRA on 1,231 signal-peptide-positive examples. Full fine-tuning led this metric at 150M and 3B.
- Used repeated runs and prior controls to test whether promising settings held up; two single-seed settings reversed on three-seed confirmation.

## Research Manuscript

D. S. Abubaker, M. Faheem, **A. S. Mukherjee**, S. Othman, S. Sifar, S. Sthapit, Y. Wang, M. Shao, and M. Shafique.
*Bio-LoRA: Biological Priors as Inductive Bias for Low-Rank Adaptation.*
Unpublished manuscript, 2026. Co-first author.
Contribution and follow-up experiments

## Independent Research

**Synthetic Pilot Lab, Independent Capstone** *Oct 2026 - present*
*In development; first diagnostic specified*

- Formulated a diagnostic study of whether synthetic participants can expose faults in experimental software before human piloting.
- Specified paired correct and faulty versions of a contribution game, scripted trajectories that expose or avoid an accounting fault, and independent checks of expected states and payoffs.

## Selected Projects

**FieldLens, Evidence-linked Incident Review** *Oct 2026*

- Directed development of a local-model incident-review prototype using fictional records; specified retrieval limits, exact source references, and a deterministic fallback for inference failures.
- Defined evaluation criteria for comparing free-prose and constrained-selection answers on matched synthetic cases: validity, relevance, completeness, and inference failures.

**Hinglish Named-Entity Recognition** *Fall 2024*

- Contributed training pipelines, evaluation tooling, and error analysis in a four-person project using mBERT and XLM-RoBERTa on COMI-LINGUA.
- Evaluated encoders on the full 4,829-example test split and zero-shot language-model baselines on a separate 100-example sample.

**CAMP, Campus Asset Management Platform** *Fall 2024*

- Built staff-workflow REST endpoints, alerts, and dashboard statistics in a five-person React, Express, and MongoDB project; configured backend integration testing and client API utilities.
- Served as Product Owner and Scrum Master in rotating team sprints.

## Applied Experience

**Machine Learning Engineer Intern, Zeek** *Jun - Aug 2025*

- Built customer-segmentation workflows and compared preprocessing and model choices; used SHAP to inspect feature contributions.

**Market Research Consultant, Global Media Alliance Broadcasting** *Jun 2024*

- Conducted audience research and designed surveys; presented findings to senior leadership.

## Service

**First Year Program Facilitator, NYU Abu Dhabi** *Aug - Dec 2024*
Facilitated workshops, group activities, and weekly check-ins for more than 20 first-year students.
**Front of House Staff, NYU Abu Dhabi Arts Center** *Jan - May 2024*
Supported guest check-in, ticketing, and event operations for audiences exceeding 500 attendees.

## Technical Skills

**Programming and data:** Python, JavaScript, C++, pandas, NumPy, SciPy
**Machine learning:** PyTorch, Transformers, scikit-learn, LoRA
**Systems:** FastAPI, Empirica, Slurm, pytest, Git